

Fixed-Case Certificate for $D = \{3, 4, 7\}$, $k = 1$

MathWise DI² Fixed-Case Proof Package

1 Scope and Non-Claims

This proof package concerns only the fixed case

$$D = \{3, 4, 7\}, \quad k = 1.$$

The result proved here is only

$$[582, \infty) \subset A_1(\{3, 4, 7\}).$$

No general BEGL conjecture is claimed.

No result is claimed for any other set D .

No result is claimed for $k \neq 1$.

No lower endpoint below 582 is claimed.

No optimality claim is made.

No density or asymptotic claim is made.

2 Theorem Statement

Let $A_1(\{3, 4, 7\})$ denote the set of integers representable as a subset sum of distinct powers generated by

$$3^a, \quad 4^b, \quad 7^c,$$

with positive exponents.

Theorem 1 (Fixed case $D = \{3, 4, 7\}, k = 1$).

$$[582, \infty) \subset A_1(\{3, 4, 7\}).$$

3 Initial Power List and Interval Certificate

The initial ordered power list is

$$b_1, \dots, b_{15} = 3, 4, 7, 9, 16, 27, 49, 64, 81, 243, 256, 343, 729, 1024, 2187.$$

The certified interval is

$$[582, 4460] \subset \text{SubSum}(b_1, \dots, b_{15}).$$

The fixed certificate parameters are

$$M = 582,$$

$$L_{15} = 4460 - 582 = 3878,$$

$$S_0 = 5042,$$

$$b_{15} = 2187,$$

$$b_{16} = 2401.$$

Thus the proof begins from the finite interval certificate

$$[582, 4460] \subset \text{SubSum}(b_1, \dots, b_{15}).$$

4 Direct Interval-Overlap Lemma

Lemma 1 (Interval-overlap propagation). *Let*

$$[M, N] \subset \text{SubSum}(S)$$

for a finite set of distinct allowed terms S . Let p be a new unused allowed term. If

$$p \leq N - M + 1,$$

then

$$[M, N + p] \subset \text{SubSum}(S \cup \{p\}).$$

Proof. The interval $[M, N]$ is already covered. Adding p to each representation covers

$$[M + p, N + p].$$

If

$$p \leq N - M + 1,$$

then

$$M + p \leq N + 1,$$

so the intervals $[M, N]$ and $[M + p, N + p]$ overlap or touch. Therefore their union covers

$$[M, N + p].$$

Hence

$$[M, N + p] \subset \text{SubSum}(S \cup \{p\}).$$

□

Let B be the ordered set of powers generated by 3, 4, 7:

$$B = \{3^a : a \geq 1\} \cup \{4^b : b \geq 1\} \cup \{7^c : c \geq 1\}.$$

For $t \in B$, define

$$S(t) = \sum_{\substack{p \in B \\ p < t}} p.$$

For a future ordered power $t > 2187$, the direct interval-overlap condition is

$$S(t) - t \geq 1163.$$

Equivalently, if $b_j = t$ is the next future term, then

$$b_j \leq L_{j-1} + 1.$$

5 Weighted Next-Gap Identities

For any future power $t > 2187$, define the next-power gaps

$$\Delta_3(t) = \min\{3^a : 3^a > t\} - t,$$

$$\Delta_4(t) = \min\{4^b : 4^b > t\} - t,$$

$$\Delta_7(t) = \min\{7^c : 7^c > t\} - t.$$

Only the two competing bases are used in each case.

Lemma 2 (Weighted next-gap identities). *The following identities hold.*

If $t = 3^r$, $r \geq 8$, then

$$S(t) - t - 1163 = \frac{2\Delta_4 + \Delta_7 - 7002}{6}.$$

If $t = 4^s$, $s \geq 6$, then

$$S(t) - t - 1163 = \frac{3\Delta_3 + \Delta_7 - 7002}{6}.$$

If $t = 7^q$, $q \geq 4$, then

$$S(t) - t - 1163 = \frac{3\Delta_3 + 2\Delta_4 - 7002}{6}.$$

Proof. For $t = 3^r$, let

$$4^a = 3^r + \Delta_4, \quad 7^b = 3^r + \Delta_7$$

be the least larger 4- and 7-powers. Then

$$S(3^r) = \frac{3^r - 3}{2} + \frac{4^a - 4}{3} + \frac{7^b - 7}{6}.$$

Substituting $4^a = 3^r + \Delta_4$ and $7^b = 3^r + \Delta_7$ gives

$$S(3^r) - 3^r - 1163 = \frac{2\Delta_4 + \Delta_7 - 7002}{6}.$$

The cases $t = 4^s$ and $t = 7^q$ are identical algebraic substitutions. For $t = 4^s$, use

$$3^a = 4^s + \Delta_3, \quad 7^b = 4^s + \Delta_7.$$

For $t = 7^q$, use

$$3^a = 7^q + \Delta_3, \quad 4^b = 7^q + \Delta_4.$$

This yields the stated identities. □

6 Weighted Gap Lemma

Lemma 3 (Weighted Gap Lemma). *For every future power $t > 2187$ generated by 3, 4, 7, the corresponding weighted next-power gap satisfies:*

If $t = 3^r$, $r \geq 8$, then

$$2\Delta_4 + \Delta_7 \geq 7002.$$

If $t = 4^s$, $s \geq 6$, then

$$3\Delta_3 + \Delta_7 \geq 7002.$$

If $t = 7^q$, $q \geq 4$, then

$$3\Delta_3 + 2\Delta_4 \geq 7002.$$

Proof. The proof is by finite full paired modular certificates. The full paired modular certificates are supplied in `WEIGHTED_GAP_CERTIFICATE_TABLES.md` and replayed by `verify_weighted_gap_certificates.py`.

A failure of the first inequality gives a bounded full system

$$\begin{aligned} 4^a - 3^r &= u, \\ 7^b - 3^r &= v, \\ 2u + v &\leq 7001, \\ r &\geq 8, \end{aligned}$$

with

$$1 \leq u \leq 3500, \quad 1 \leq v \leq 7000.$$

A failure of the second inequality gives

$$\begin{aligned} 3^a - 4^s &= u, \\ 7^b - 4^s &= v, \\ 3u + v &\leq 7001, \\ s &\geq 6, \end{aligned}$$

with

$$1 \leq u \leq 2333, \quad 1 \leq v \leq 7000.$$

A failure of the third inequality gives

$$\begin{aligned} 3^a - 7^q &= u, \\ 4^b - 7^q &= v, \\ 3u + 2v &\leq 7001, \\ q &\geq 4, \end{aligned}$$

with

$$1 \leq u \leq 2333, \quad 1 \leq v \leq 3500.$$

The stabilized residue filters are as follows.

For Case A, $r \geq 8$, so

$$3^r \equiv 0 \pmod{6561}.$$

Hence

$$\begin{aligned} 4^a &\equiv u \pmod{6561}, \\ 7^b &\equiv v \pmod{6561}. \end{aligned}$$

Use

$$\text{ord}_{6561}(4) = 2187, \quad \text{ord}_{6561}(7) = 2187.$$

For Case B, $s \geq 6$, so

$$4^s \equiv 0 \pmod{4096}.$$

Hence

$$3^a \equiv u \pmod{4096},$$

$$7^b \equiv v \pmod{4096}.$$

Use

$$\text{ord}_{4096}(3) = 1024, \quad \text{ord}_{4096}(7) = 512.$$

For Case C, $q \geq 4$, so

$$7^q \equiv 0 \pmod{2401}.$$

Hence

$$3^a \equiv u \pmod{2401},$$

$$4^b \equiv v \pmod{2401}.$$

Use

$$\text{ord}_{2401}(3) = 2058, \quad \text{ord}_{2401}(4) = 1029.$$

The attached replay verifier reconstructs the residue-admissible systems and confirms

$$\text{Case A remaining} = 0, \quad \text{Case B remaining} = 0, \quad \text{Case C remaining} = 0.$$

The certificate checks the full paired systems, not only the difference equation. Thus every possible failure system is excluded, and the three weighted inequalities hold. $\square$

7 Full Paired Modular Certificates

The certificate method eliminates full paired systems. The compressed certificate table and replay verifier are external-review attachments to this proof package.

Case A

The full system is

$$4^a - 3^r = u,$$

$$7^b - 3^r = v,$$

$$r \geq 8.$$

After the stabilized residue filter modulo 6561, there are

$$1,361,889$$

residue-admissible systems.

They are eliminated by full paired modular certificates using the following moduli:

$$5, 13, 19, 31, 37, 41, 43, 73, 109, 487, 2917, 17497.$$

The final Case A remaining row before the expanded modulus is

$$(u, v) = (1909, 214),$$

with forced classes

$$a \equiv 1464 \pmod{2187},$$

$$b \equiv 733 \pmod{2187}.$$

It is eliminated modulo

$$17497.$$

Case B

The full system is

$$\begin{aligned}3^a - 4^s &= u, \\7^b - 4^s &= v, \\s &\geq 6.\end{aligned}$$

After the stabilized residue filter modulo 4096, there are

$$255,938$$

residue-admissible systems.

They are eliminated by full paired modular certificates using the following moduli:

$$5, 11, 13, 17, 19, 29, 31, 37, 41, 43, 257.$$

Case C

The full system is

$$\begin{aligned}3^a - 7^q &= u, \\4^b - 7^q &= v, \\q &\geq 4.\end{aligned}$$

After the stabilized residue filter modulo 2401, there are

$$1,501,666$$

residue-admissible systems.

They are eliminated by full paired modular certificates using the following moduli:

$$5, 13, 17, 19, 29, 31, 37, 41, 43, 73, 197, 1373.$$

The final two Case C remaining rows before modulo 1373 are

$$(u, v) = (386, 681),$$

$$(u, v) = (1844, 681).$$

Both are eliminated modulo

$$1373.$$

Least-Larger Audit

The full paired modular certificates eliminate stronger systems than the least-larger next-power systems. Therefore the least-larger inequalities are not needed in the final modular proof.

8 Direct Propagation

By the Weighted Gap Lemma and the weighted next-gap identities, for every future ordered power

$$t > 2187$$

generated by 3, 4, 7,

$$S(t) - t \geq 1163.$$

Equivalently, for every future ordered power b_j ,

$$b_j \leq L_{j-1} + 1.$$

Starting from

$$[582, 4460] \subset \text{SubSum}(b_1, \dots, b_{15}),$$

the interval-overlap lemma applies successively to every later ordered power.

Therefore the covered interval propagates indefinitely:

$$[582, \infty) \subset A_1(\{3, 4, 7\}).$$

9 Final Conclusion

Thus, for the fixed case

$$D = \{3, 4, 7\}, \quad k = 1,$$

we conclude

$$[582, \infty) \subset A_1(\{3, 4, 7\}).$$

No general BEGL result is claimed.

No result for other D is claimed.

No result for $k \neq 1$ is claimed.

No endpoint below 582 is claimed.

10 Appendix: Survivor-System Compatibility

The survivor-system framework was separately closed during the fixed-case analysis.

The recorded survivor-system closure status is:

Case 1: closed,

Case 2: closed,

Case 3: closed.

The augmented coarse candidate closure status is:

Case 1: 51/51 closed,

Case 2: 52/52 closed,

Case 3: 38/38 closed.

This survivor-system closure remains compatible with the fixed-case result.

However, the final proof above does not require the survivor-system framework, because the direct interval-overlap route supplies an independent certificate path:

finite interval certificate+interval-overlap lemma+Weighted Gap Lemma $\implies [582, \infty) \subset A_1(\{3, 4, 7\})$.